

Dear Sir or Madam,

I am applying for the Senior Software Engineer at Terra Quantum because I want to continue growing as a backend engineer working on large scale, data-intensive systems.

As Co-founder and CTO of WD AI, I spent more than a year designing and running a full-stack AI platform that scaled to over 200,000 users and around 900 concurrent sessions in under four months. This role required me to take full ownership of infrastructure, reliability, and scalability, and it gave me hands-on experience with cloud-native systems and production operations. I built microservices in NestJS, designed REST APIs, and deployed containerized services using Docker and Kubernetes. I ran GPU workloads in the cloud, used RabbitMQ for job queues, Redis and PostgreSQL for data storage and cache, and maintained CI/CD pipelines for reliable releases. For monitoring, I relied on centralized logging, Prometheus, and Grafana to track performance and system health. The entire frontend was built in React, and in total I have more than two years of full-stack experience in real-world environments. Across these roles, I worked on real-time data streaming and state synchronization in a multi-user 3D editor collaboration platform at Huawei, using SignalR to deliver low-latency updates.

For the past two years I have been working at a scientific startup QC Design on a Python simulation library for quantum computing. I set a high bar for the tempo of delivery and the quality of the code. I worked on the core infrastructure of the library achieving over 10,000x speedup by implementing CPython bindings for the C++ code. I also worked on implementing new algorithms for the codes and simulators from papers, verifying them against the literature. I used GCP to run large-scale quantum simulations, generate substantial datasets, and build analysis pipelines.

All of this experience has shaped my focus on building reliable backend systems, and I am eager to deepen it further at Terra Quantum!

Yours truly,

Ivan Ogloblin